26 05 21-CONDUCTOR CONNECTIONS AND DEVICES

PART 1 – GENERAL

1.01 SECTION INCLUDES

A. Conductor connections and devices.

PART 2 – PRODUCTS

2.01 ACCEPTABLE MANUFACTURERS

A. Conductor connectors for general wiring:

1. Size Nos. 6 or larger, copper:
    a. Ilsco Corp.
    b. Thomas & Betts Co.
    c. Burndy Corp.
    d. Square D Versa-Crimp.
2. Size Nos. 1/0 or larger, aluminum:
    a. Thomas Betts Co.
    b. Square D Versa-Crimp.

B. Insulating tape:

1. Minnesota Mining & Mfg. Co. (3M).
2. Plymouth Rubber Co.

2.02 MATERIAL AND FABRICATION

A. Conductor connectors for general wiring:

1. Splice and terminate with compression or pressure type connectors and terminal lugs.
2. Outdoor or wet locations: Silicon filled safety connectors by King Technology, Inc. or Owner-approved equivalent.

B. Multi-conductor cables:

1. Ring tongue terminal lug with insulated grip and insulation support:

C. Connector sealing packs for splices that require complete protection from dampness and water.

1. Scotchlok Nos. 3576, 3577 and 3578, by 3M Company.

D. Electrical insulating resin: Scotchcast Electrical Insulating Resin 400 by 3M Company.

PART 3 – EXECUTION

3.01 INSTALLATION

A. Terminate and splice feeder conductors as follows:

1. Strip insulation by penciling or paring very carefully so as not to damage the conductor.
2. Remove enough insulation so that conductor can be fully inserted into connector sleeve or terminal lug.
3. Apply oxide inhibitor recommended by connector manufacturer to all exposed conductors and wire brush to fully penetrate spaces between strands.
4. Use hydraulic type compression tool: Thomas & Betts Type TBM15 or Square D Versa-Crimp.
5. Operate pump until bypass has been reached.
6. Place full amount of crimps on connectors and terminal lugs as recommended by manufacturer.
7. Use compression type connector and terminal lugs for splices and terminations.
8. Apply torque values specified by terminal lug manufacturer.
9. Use "breakaway" type torque wrench to prevent over tightening.

B. Insulate splices with plastic electrical tape: Scotch No. 33+ or Tomic No. 1T.

C. Terminate all control wires with terminal lugs on barriered terminal boards. If splices are needed, use same procedure, installing terminal board in a junction box for protection.

D. Provide lugs for conductors terminating on secondary side of substation.

END OF SECTION 26 05 21